M2-Thesis

Daniel Gagliardi

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Preface

Throughout my undergraduate and graduate studies in mathematics I have been caught between two main interests: the applied world of machine learning, and the theoretical world of topology and geometry. While this is often viewed as an unusual mix to choose, I have always found it greatly enjoyable.

This culminated in me wanting to answer the question of what is happening to the "shape" of data throughout a machine learning process, and also what even is the "shape" of data.

The value in understanding this is twofold: first, to be able to make better models by leveraging our understanding of how they work, and second, to be able to understand how to interpret what these models are doing. The dance between naturally arising data and the model trying to fit without knowing how the data is generated is a curious one. My main motivation was to try to understand this.

One big idea is the notion of data lying on some low-dimensional space, which makes both learning and sampling from distributions supported on this lower-dimensional space easier. However this belief only truly simplifies the system in the small noise regime. All sorts of complex geometry can happen in the higher noise regimes. We do not, as we will see, have a nicely behaved interpolating family of manifolds.

Before doing the IASD M2 program, I had written a master's dissertation on diffusion models under the manifold hypothesis, under the supervision of George Deligiannidis. During this I became deeply interested in dimension estimation of datasets. This at best vauge and at worse terribly ill-posed question seemed to me one where and answer would tell us a lot about the natural geometry of data, and how diffusion models are able to make use of this. Almost every paper I found on the interplay between manifolds and diffusion models cited work from Eddie Aamari.

In November 2025 I reached out to Eddie Aamari to further discuss what the geometry of data could look like, and this led to this internship project. The initial goal was to understand the interplay between geometry affecting sampling error and how this could encourage simpler embeddings in a latent diffusion setting. However, given the complexity of work being done limited time was spent on investigating autoencoders so far.

A key challenge of this project has been stirking the balance between geometry: where things are well defined, and diffusion of probability densities: where simply by adding noise we violently change the notions of geometry we would like to equip our density with. This tradeoff has led to looking at several objects, which while interesting in their own right have not been helpful in bounding sampling error. We do think the final part on posterior covariance fields section 6 does provide new insight on where true complexity lies in the diffusion process.

Abstract

This is the abstract of the thesis.

Algebra is the offer made by the devil to the mathematician. The devil says: 'I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.'

— ***Michael Atiyah***

1 Introduction

1.1 Initial Project

The progress in generative modelling over the last decade has been rapid. A key shift was the introduction of score based generative models (SGM), also known as diffusion models, which allow us to interpolate between a know distribution (usually a standard Gaussian), and a target distribution p_{data}, by training a network based on data $X \sim p_{\text{data}}$. By repeatedly adding noise to a target density, and training a denoising network, one can sample from p_{data}. Now, flow matching methods are the standard implementation of diffusion based generative models. These crucially can be viewed as a SGM.

While vast amounts of work has been done on improving the capabilities of diffusion models, the goemetry of the entire diffusion process has been under explored.

Initially, this research internship aimed to investigate the geometry of both the autoencoder and denoiser in latent diffusion models. If one can show that the geometry of a target distribtuion influences the estimation of scores, discretisation error, or sampling stability, then the case can be made that when the autoencoder and diffusion model are trained jointly, that the autoencoder learns a more geometrically regular embedding.

We spent the internship so far almost entirely looking at the geometry of diffusion.

1.2 Existing work

There exists literature on how different geometric aspects affect the diffusion process, but these are mostly asymptotic results for highly regular target densities. [list some papers!!!]

The main challenge of the question "how does geomerty affect diffusion" is what do we mean by geometry? There are some issues with existing methods which make use of dimension and reach/curvature of the support of p_{data} :

- Dimension:
  - Ill-posed to assign a dimension $d < D$ to a density which is absolutely continuous with respect to $\mathbb{R}^D$
  - Dimension of p_{data} may not be constant
  - Dimension of p_{data} may not be well defined everywhere, for example at intersections
  - Only considers dimension of target, now how effective dimension might change.
- Reach, curvature:
  - Can be $0, \infty$ at intersections or corners
  - Can vary wildy along the support of p_{data}

1.3 Objects Investigated

This gap in the literature is the primary of the project. As such, a new object other than dimension or reach or curvature had to be found. Throughout we investigated several before landing on the final and most promising object: the posterior covariance.

- Pullback metric of flow $G_{t,T}(x)$
- Level sets
- Ridges
- Posterior Covariance

[!!!make a table comparing them]

1.4 Main Contribution

We find that the spectrum of the posterior covariance captures geometric variation, and use functionals of the spectrum to understand geometric variation throughout the diffusion process. One of the things this allows us to do is pick a disctetisation schedule to reduce the discretisation error in sampling.

2 Background

Here we go over the necessary background needed for the project.

2.1 The Manifold Hypothesis

The manifold hypothesis is an explaination of the effectiveness of machine learning for unstructured high-dimensional data. It states that a distribution $X \sim p_{\text{data}}, X \in \mathbb{R}^D$ lies on (or is concentrated around) a submanifold $\mathcal{M}$ of lower dimension $d \ll D$. This effectively makes the number of degrees of freedom d, allowing models to learn these lower dimensional spaces in settings where we may have $n < D$ data points.

The main argument for the manifold hypothesis is the curse of dimensionality, whereby there are too many dimensions making data very sparse and hard for models to learn from. Many counterintuitve behaviours are exhibited in high dimensional space, for example standard Gaussian data from $\mathcal{N}(0; \mathrm{Id}_D)$ concentrating around the shell of radius $\sqrt{D}$.

The justification of the manifold hypothesis is that it is a good canidate model for naturaly arising data, which tends to be locally continuous, and have fewer degrees of freedom. The space of natural images is far smaller than the space of all possible images, for example.

There are variations of the manifold hypothesis, which make it less restrictive an assumption. These include:

- The union of manifolds hypothesis Brown et al. (2022a), Brown et al. (2022b), which argue that standard manifold hypothesis is incomplete, and a union of manifolds of potentially different dimensions is a more realistic assumption
- The stratified space hypothesis Aamari and Berenfeld (2024), which extends this idea but allow for intersections between manifolds of different dimension

2.2 Flow Matching as a Score Based Generative Model

Score Based Generative models (SGMs) Ho et al. (2020), Song and Ermon (2019), Song et al. (2020b) introduced the notion of generative modelling by estimating gradients of the density. In fact, in the abstract of Song and Ermon (2019) the authors state "Because gradients can be ill-defined and hard to estimate when the data resides on **low-dimensional manifolds**, we perturb the data with different levels of Gaussian noise, and jointly estimate the corresponding scores, i.e., the vector fields of gradients of the perturbed data distribution for all noise levels."

Since the introduction of SGMs, they have been refined into their canonical form: flow matching models Lipman et al. (2022), Albergo et al. (2025). Here we give a brief overview.

2.2.1 Gaussian Flows and Score-Based Flow Matching

In generative modeling via flow matching, we aim to transform a base noise distribution $p_1 = \mathcal{N}(0, I_d)$ into a complex target data distribution p_0. Let $X_0 \sim p_0$ and $X_1 = \varepsilon \sim \mathcal{N}(0, I_d)$. The linear conditional flow $X_t = \varphi_t(X_0, \varepsilon)$ interpolates between data and noise according to:

$$X_t = c_t X_0 + \sigma_t \varepsilon$$

where c_t, σ_t are differentiable boundary-satisfying schedules with $c_0 = 1, \sigma_0 = 0$ and $c_1 = 0, \sigma_1 = 1$. Differentiating with respect to time yields the conditional vector field:

$$\dot{X}_t = \dot{c}_t X_0 + \dot{\sigma}_t \varepsilon$$

Vector Field Formulation The marginal probability density p_t of X_t evolves according to the continuity equation $\partial_t p_t = -\nabla \cdot (v_t p_t)$. The unconditional ODE trajectory $\dot{x}_t = v_t(x_t)$ generates these exact same marginals p_t when initialized at $x_0 \sim p_0$. The marginal velocity field $v_t(x)$ is given by the conditional expectation:

$$v_t(x) = \mathbb{E}\left[\dot{X}_t \mid X_t = x\right] = \dot{c}_t \mathbb{E}\left[X_0 \mid X_t = x\right] + \dot{\sigma}_t \mathbb{E}\left[\varepsilon \mid X_t = x\right]$$

Derivation via Tweedie's Formula Because p_t is the distribution of $c_t X_0$ corrupted by Gaussian noise $\sigma_t \varepsilon \sim \mathcal{N}(0, \sigma_t^2 I_d)$, Tweedie's denoising identity provides closed-form expressions for the conditional expectations in terms of the score function $\nabla \ln p_t(x)$:

$$\mathbb{E}\left[\varepsilon \mid X_t = x\right] = -\sigma_t \nabla \ln p_t(x)$$

$$\mathbb{E}\left[X_0 \mid X_t = x\right] = \frac{x + \sigma_t^2 \nabla \ln p_t(x)}{c_t}$$

Substituting these expectations into the expression for $v_t(x)$ yields:

$$\begin{aligned}
v_t(x) &= \dot{c}_t \left(\frac{x + \sigma_t^2 \nabla \ln p_t(x)}{c_t} \right) + \dot{\sigma}_t \left(-\sigma_t \nabla \ln p_t(x) \right) \\
&= \frac{\dot{c}_t}{c_t} x + \left(\frac{\dot{c}_t \sigma_t^2}{c_t} - \dot{\sigma}_t \sigma_t \right) \nabla \ln p_t(x) \\
&= \frac{\dot{c}_t}{c_t} x + \sigma_t^2 \left[\frac{\dot{c}_t}{c_t} - \frac{\dot{\sigma}_t}{\sigma_t} \right] \nabla \ln p_t(x)
\end{aligned}$$

Implications and Equivalence The velocity field $v_t(x)$ is completely determined by the schedules c_t, σ_t and the score function $\nabla \ln p_t(x)$. The parameterized target models are related by linear transformations:

$$\nabla \ln p_t(x) \quad \Longleftrightarrow \quad \mathbb{E}\left[\varepsilon \mid X_t = x\right] \quad \Longleftrightarrow \quad \mathbb{E}\left[X_0 \mid X_t = x\right] \quad \Longleftrightarrow \quad v_t(x)$$

Learning any one of these targets allows immediate recovery of the others. Sampling via the flow ODE $\dot{x}_t = v_t(x_t)$ using a score model $\nabla \ln p_t$ is mathematically equivalent to running the probability flow (DDIM Song et al. (2020a)) ODE in diffusion models.

3 Anisotropic Stability Bound

We record a stability estimate for generator matching that keeps track of *which* directions of a drift error are amplified and which are damped. The scalar one-sided Lipschitz estimate propagates every error with a single worst-case rate; it is sharp only in an isotropic sense. The estimate below replaces that scalar rate by the derivative flow of the learnt dynamics, a matrix that sees the direction of the error. We isolate the mechanism in the pure-transport case first: throughout Sections 3.1–3.3 the diffusion coefficient is set to zero, and the additive-noise case is recovered verbatim by averaging the derivative flow over the Brownian trajectories (Remark 3.3).

3.1 The deterministic transport mechanism

Consider two time-dependent velocity fields $v_t, \hat{v}_t : \mathbb{R}^D \to \mathbb{R}^D$ for $0 \le t \le T$, smooth enough in space and integrable in time that their characteristic flows are globally defined and differentiable in the initial condition:

$$\tfrac{\mathrm{d}}{\mathrm{d}s}\Phi_{t,s}(x) = v_s\big(\Phi_{t,s}(x)\big), \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\hat{\Phi}_{t,s}(x) = \hat{v}_s\big(\hat{\Phi}_{t,s}(x)\big), \qquad \Phi_{t,t} = \hat{\Phi}_{t,t} = \mathrm{Id}. \tag{1}$$

We write $\delta v_t := v_t - \hat{v}_t$ for the velocity error and think of v_t as the exact field and $\hat{v}_t$ as the learnt one.

The estimate is built entirely from the derivative of the *learnt* flow,

$$\hat{J}_{t,T}(x) := D_x\hat{\Phi}_{t,T}(x), \qquad \tfrac{\mathrm{d}}{\mathrm{d}s}\hat{J}_{t,s}(x) = \nabla\hat{v}_s\big(\hat{\Phi}_{t,s}(x)\big)\,\hat{J}_{t,s}(x), \quad \hat{J}_{t,t}(x) = \mathrm{Id}. \tag{2}$$

Thus $\hat{J}_{t,T}(x)h$ is the terminal displacement at time T produced by an infinitesimal perturbation h applied at (t,x) and transported by the learnt dynamics. The learnt flow acts as a magnifying glass on errors: an error committed at time t is not scaled by a worst-case constant but carried by $\hat{J}_{t,T}$ along its own direction. The following identity makes this exact.

Lemma 3.1 (Telescoping identity along characteristics)**.** *For every initial point $x_0 \in \mathbb{R}^D$, with $X_t := \Phi_{0,t}(x_0)$ the exact characteristic,*

$$\Phi_{0,T}(x_0) - \hat{\Phi}_{0,T}(x_0) = \int_0^T \hat{J}_{t,T}(X_t)\,\delta v_t(X_t)\,\mathrm{d}t. \tag{3}$$

Proof. Define $F(t) := \hat{\Phi}_{t,T}(X_t)$, which follows the exact dynamics up to time t and the learnt dynamics from t to T. Its endpoints are the fully learnt and fully exact terminal points, $F(0) = \hat{\Phi}_{0,T}(x_0)$ and $F(T) = \Phi_{0,T}(x_0)$. Differentiating in t,

$$\tfrac{\mathrm{d}}{\mathrm{d}t}F(t) = \partial_t\hat{\Phi}_{t,T}(X_t) + D\hat{\Phi}_{t,T}(X_t)\,\dot{X}_t.$$

Starting the learnt flow an instant later removes one short learnt step, so $\partial_t\hat{\Phi}_{t,T}(x) = -D\hat{\Phi}_{t,T}(x)\,\hat{v}_t(x)$. Together with $\dot{X}_t = v_t(X_t)$ and $D\hat{\Phi}_{t,T}(X_t) = \hat{J}_{t,T}(X_t)$ this gives

$$\tfrac{\mathrm{d}}{\mathrm{d}t}F(t) = \hat{J}_{t,T}(X_t)\big(v_t(X_t) - \hat{v}_t(X_t)\big) = \hat{J}_{t,T}(X_t)\,\delta v_t(X_t).$$

Integrating $F(T) - F(0)$ from 0 to T yields (3). $\square$

Identity (3) is the heart of the estimate. The error made at each time enters the terminal gap through the derivative flow evaluated in the actual direction of that error, with no worst-case constant and no loss.

3.2 From the telescoping identity to the matrix bound

We turn the pathwise identity into a bound on the terminal laws by coupling. Let $X_0 \sim \rho_0$ and transport it by both flows, $X_t := \Phi_{0,t}(X_0)$ and $\hat{X}_t := \hat{\Phi}_{0,t}(X_0)$, with terminal laws $\rho_T := \mathrm{Law}(X_T)$ and $\hat{\rho}_T := \mathrm{Law}(\hat{X}_T)$. Running both flows from the same X_0 is a coupling of ρ_T and $\hat{\rho}_T$, so $W_1(\rho_T, \hat{\rho}_T) \leq \mathbb{E}\|X_T - \hat{X}_T\|$. Applying Lemma 3.1 pathwise at $x_0 = X_0$ and the triangle inequality,

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \mathbb{E}\big\|\hat{J}_{t,T}(X_t)\,\delta v_t(X_t)\big\|\,\mathrm{d}t = \int_0^T \int_{\mathbb{R}^D} \big\|\hat{J}_{t,T}(x)\,\delta v_t(x)\big\|\,\rho_t(\,\mathrm{d}x)\,\mathrm{d}t, \tag{4}$$

where the last step uses $X_t \sim \rho_t$. It remains only to name the quadratic form hidden in the norm.

Theorem 3.2 (Anisotropic W_1 stability)**.** *Define the pullback amplification metric*

$$G_{t,T}(x) := \hat{J}_{t,T}(x)^\top \hat{J}_{t,T}(x), \qquad \textit{so that} \qquad \|\hat{J}_{t,T}(x)h\|^2 = \big\langle h,\, G_{t,T}(x)\,h\big\rangle. \tag{5}$$

Then

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \int_{\mathbb{R}^D} \big\langle \delta v_t(x),\, G_{t,T}(x)\,\delta v_t(x)\big\rangle^{1/2}\,\rho_t(\,\mathrm{d}x)\,\mathrm{d}t. \tag{6}$$

Proof. Substitute the identity in (5) into (4). $\square$

The matrix $G_{t,T}(x)$ is the Euclidean metric at the terminal time pulled back to the tangent space at (t, x) by the learnt flow: it lives at time t but measures squared lengths after transport to T. Its eigenvectors are the input directions at (t, x) and its eigenvalues are the squared stretching factors of those directions. In the deterministic case (5) is an exact identity, so no Cauchy–Schwarz slack is incurred in passing from the derivative flow to the quadratic form. The bound (6) is anisotropic precisely because it evaluates $G_{t,T}$ on the actual error direction $\delta v_t(x)$ rather than on the worst possible direction.

Remark 3.3 (Additive diffusion). When both dynamics carry the same additive diffusion coefficient, the Brownian term has no derivative in the initial condition, so the learnt derivative flow $\hat{J}_{t,s}$ still solves the linear equation (2) along each trajectory. The metric becomes the trajectory average $G_{t,T}(x) = \mathbb{E}\big[\hat{J}_{t,T}(x)^\top \hat{J}_{t,T}(x)\big]$, and Jensen's inequality turns the exact identity in (5) into the upper bound $\mathbb{E}\|\hat{J}_{t,T}(x)\delta v_t(x)\| \leq \langle \delta v_t, G_{t,T}\delta v_t\rangle^{1/2}$, so (6) holds unchanged. The deterministic case treated here is the load-bearing one.

3.3 Recovering the one-sided Lipschitz estimate

The classical scalar estimate is the isotropic specialization of Theorem 3.2, obtained by discarding the direction of the error. Let $S_t(x) := \frac{1}{2}\big(\nabla \hat{v}_t(x) + \nabla \hat{v}_t(x)^\top\big)$ be the symmetric part of the learnt Jacobian, and suppose the learnt field is one-sided Lipschitz with rate ℓ_t, meaning

$$\lambda_{\max}\big(S_t(x)\big) \le \ell_t \qquad \text{for all } x. \tag{7}$$

The condition constrains expansion only: it bounds the largest rate at which two trajectories separate, and places no bound on contraction or on the rotational part of $\nabla \hat{v}_t$.

Corollary 3.4 (Scalar one-sided Lipschitz bound)**.** *Under* (7)*,*

$$G_{t,T}(x) \;\preceq\; \exp\Big(2\int_t^T \ell_s \,\mathrm{d}s\Big)\,\mathrm{Id}, \tag{8}$$

and consequently

$$W_1(\rho_T, \hat{\rho}_T) \le \int_0^T \exp\Big(\int_t^T \ell_s \,\mathrm{d}s\Big)\,\|\delta v_t\|_{L^1(\rho_t)}\,\mathrm{d}t. \tag{9}$$

Proof. Fix a direction h and set $z_s := \hat{J}_{t,s}(x)h$, which solves $\dot{z}_s = \nabla \hat{v}_s(\hat{\Phi}_{t,s}(x))\,z_s$ by (2). Only the symmetric part contributes to the growth of the norm, so

$$\tfrac{\mathrm{d}}{\mathrm{d}s}\|z_s\|^2 = 2\big\langle z_s,\, S_s(\hat{\Phi}_{t,s}(x))\,z_s\big\rangle \le 2\ell_s\|z_s\|^2.$$

Grönwall's lemma gives $\|\hat{J}_{t,T}(x)h\| \le \exp\big(\int_t^T \ell_s\,\mathrm{d}s\big)\|h\|$, which is (8). Inserting (8) into (6) and using $\langle \delta v_t, G_{t,T}\delta v_t\rangle^{1/2} \le \exp(\int_t^T \ell_s\,\mathrm{d}s)\|\delta v_t\|$ yields (9). □

Corollary 3.4 exhibits the scalar estimate as the Loewner domination (8): the matrix $G_{t,T}$ is replaced by its largest eigenvalue, uniformly over directions. The bound is correct, but it forgets the eigenstructure of $G_{t,T}$, and in particular treats a strongly contracted error direction as though it were the most expansive one. The next subsection makes this gap concrete in a two-direction model, where a normal error is carried by a contraction factor that the scalar constant cannot see.

3.4 A two-direction picture

The gap between the matrix and scalar bounds is already visible in the simplest anisotropic model. Fix an orthogonal splitting $\mathbb{R}^D = E_\parallel \oplus E_\perp$ into a tangent-like block and a normal-like block, write $x = (x_\parallel, x_\perp)$ accordingly, and take the learnt velocity to be linear, block-diagonal, expansive on $E_\parallel$ and contractive on $E_\perp$:

$$\hat{v}_t(x_\parallel, x_\perp) = \big(\ell_t x_\parallel,\; -\gamma_t x_\perp\big), \qquad \gamma_t \ge 0. \tag{10}$$

Its Jacobian $\nabla \hat{v}_t = \mathrm{diag}(\ell_t I_\parallel, -\gamma_t I_\perp)$ is constant in space, so the derivative flow (2) integrates blockwise and the pullback metric (5) is diagonal:

$$\hat{J}_{t,T} = \begin{pmatrix} e^{\int_t^T \ell_s \, \mathrm{d}s} I_\parallel & 0 \\ 0 & e^{-\int_t^T \gamma_s \, \mathrm{d}s} I_\perp \end{pmatrix}, \qquad G_{t,T} = \begin{pmatrix} e^{2\int_t^T \ell_s \, \mathrm{d}s} I_\parallel & 0 \\ 0 & e^{-2\int_t^T \gamma_s \, \mathrm{d}s} I_\perp \end{pmatrix}. \tag{11}$$

Splitting the error as $\delta v_t = \delta v_t^\parallel + \delta v_t^\perp$ with $\delta v_t^\parallel \in E_\parallel$ and $\delta v_t^\perp \in E_\perp$, the integrand of the matrix bound (6) separates,

$$\big\langle \delta v_t,\, G_{t,T}\, \delta v_t \big\rangle^{1/2} = \Big(e^{2\int_t^T \ell_s \, \mathrm{d}s} \, \|\delta v_t^\parallel\|^2 + e^{-2\int_t^T \gamma_s \, \mathrm{d}s} \, \|\delta v_t^\perp\|^2 \Big)^{1/2}. \tag{12}$$

A tangent error is carried to the terminal time by the expansion factor $e^{\int_t^T \ell}$; a normal error is carried by the contraction factor $e^{-\int_t^T \gamma}$ and is therefore damped.

The scalar estimate cannot distinguish the two. Its rate is the largest eigenvalue of the symmetric Jacobian,

$$\lambda_{\max}\big(S_t\big) = \max\{\ell_t,\, -\gamma_t\} = \ell_t \qquad (\ell_t \geq 0), \tag{13}$$

so Corollary 3.4 propagates *both* error components with the tangent expansion factor $e^{\int_t^T \ell}$. It replaces the normal contraction $e^{-\int_t^T \gamma}$ by expansion, overestimating the contribution of a normal error by the factor $e^{\int_t^T (\ell_s + \gamma_s)\, \mathrm{d}s}$. When the normal block is strongly contracting—as it is near a terminal manifold, where $\gamma_t \sim (1-t)^{-1}$ and $e^{-\int_t^T \gamma}$ decays polynomially in $1-t$—this overestimate is the entire difference between a persistent error and a damped one. This is where the anisotropy pays.

3.5 What the estimate requires of the error

The two bounds depend on the error through different amounts of information, and this raises a natural objection. The scalar bound (9) uses only the magnitude $\|\delta v_t\|_{L^1(\rho_t)}$. The matrix bound (6) uses the quadratic form $\langle \delta v_t, G_{t,T} \delta v_t \rangle$, that is, the *direction* of δv_t relative to the eigenframe of $G_{t,T}$. Since the exact field v_t is unknown, the error $\delta v_t = v_t - \hat{v}_t$ is never observed, and its direction still less so. It is not immediate that the matrix bound is any more usable than the scalar one.

The resolution is that neither bound is an estimator: both are a priori certificates. The scalar bound is valid because it implicitly places δv_t along the most expansive eigenvector of $G_{t,T}$, which is exactly the source of its looseness. The matrix bound is a family of certificates indexed by hypotheses on where the error lies, and it becomes informative as soon as one can argue, on structural grounds, that the error concentrates in contracted directions. The direction is supplied not by measuring δv_t but by the geometry of the *target*.

Near a smooth embedded manifold this structure is explicit. The exact velocity v_t, which is fixed by the score of the noised target, admits to leading order a canonical decomposition: a rigid normal restoring field of model-independent functional form, a tangential component set by the intrinsic density on the manifold, and a moving ridge that locates the normal zero of the field. Any learnt error is then graded in the *same* frame into a normal-amplitude component, a tangential-density component, and a ridge-displacement component. The matrix bound assigns each its own weight through $G_{t,T}$: normal-amplitude errors are damped by the normal contraction factor of subsection 3.4, tangential-density errors are carried by the tangential expansion factor and persist, and ridge-displacement errors are normal but enter with the singular weight σ_t^{-2}, since a normal displacement of the ridge by Δ changes the leading normal field by Δ/σ_t^2. The estimate never requires δv_t to be known; it requires the target to furnish a fixed anisotropic frame in which any error is automatically decomposed. We construct this frame in section 6.

One assumption must be stated now, because it is used throughout. The projectors defining the frame—P_t onto the tangent block and $Q_t = I - P_t$ onto the normal block—must be a single field, determined by the target and shared across all learnt fields. Re-deriving the frame from each model's own error would reintroduce model dependence into the weights and render comparisons between models meaningless. We therefore fix P_t once, from the target geometry, and read every error in that frame; concretely, P_t is built from the spectrum of the posterior-covariance operator A_t in section 6.

4 Level Sets

as we smooth data which lies on or is concentrated around a manifold two main affects occur. Smoothing in the normal direction and smoothing and tangent direction. Smoothing in the tangent direction makes a density more uniform along a manifold. Therefore our level sets should be better behaved.

Let $p^\star : \mathbb{R}^D \to \mathbb{R}_+$ denote the initial density, and let p_σ denote its Gaussian-smoothed version,

$$p_\sigma(x) = (G_\sigma * p^\star)(x), \qquad G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right). \tag{14}$$

For a regular value c, the corresponding level set is

$$\mathcal{M}_c^\sigma := \left\{x \in \mathbb{R}^D : p_\sigma(x) = c\right\}, \qquad \nabla p_\sigma(x) \neq 0 \quad \text{for } x \in \mathcal{M}_c^\sigma. \tag{15}$$

The unit normal to $\mathcal{M}_c^\sigma$ is

$$n(x) = \frac{\nabla p_\sigma(x)}{\|\nabla p_\sigma(x)\|}. \tag{16}$$

For a tangent direction $v \in T_x\mathcal{M}_c^\sigma$, the derivative of the normal is

$$D_v n = \frac{1}{\|\nabla p_\sigma\|} \left(I - nn^\top\right) D^2 p_\sigma \, v = \frac{1}{\|\nabla p_\sigma\|} P_T D^2 p_\sigma \, v, \tag{17}$$

where

$$P_T := I - nn^\top \tag{18}$$

is the orthogonal projection onto the tangent space $T_x\mathcal{M}_c^\sigma$. Thus, the variation of the normal field—and hence the local curvature of the level set—is determined by the Hessian of the density.

Gaussian smoothing regularizes not only the density itself, but all of its spatial derivatives. In particular, for any $k \geq 0$,

$$D^k p_\sigma = D^k(G_\sigma * p^\star) = (D^k G_\sigma) * p^\star. \tag{19}$$

Hence increasing σ smooths the density and simultaneously suppresses fine-scale variation in its gradient, Hessian, and higher-order derivatives.

This smoothing has a particularly simple interpretation in the Fourier domain. Writing $\widehat{p_\sigma}$ for the Fourier transform,

$$\widehat{p_\sigma}(\xi) = \widehat{G}_\sigma(\xi)\, \widehat{p^\star}(\xi) = \exp\left(-\frac{\sigma^2\|\xi\|^2}{2}\right) \widehat{p^\star}(\xi). \tag{20}$$

Thus a Fourier mode of spatial frequency $\|\xi\|$ is attenuated by $\exp(-\sigma^2\|\xi\|^2/2)$: high-frequency components decay quadratically faster in frequency. Consequently, Gaussian smoothing preferentially removes the fine-scale spatial variation of $p^\star$.

The effect on the geometry of a level set follows directly from (17). Its shape operator is

$$S := -P_T D n P_T = -\frac{1}{\|\nabla p_\sigma\|} P_T D^2 p_\sigma P_T, \tag{21}$$

up to the choice of orientation for the normal. The eigenvalues of S are the principal curvatures of $\mathcal{M}_c^\sigma$. Therefore, Gaussian smoothing affects the shape of the level set through two quantities: the Hessian $D^2 p_\sigma$, which controls variation of the normal, and $\|\nabla p_\sigma\|$, which sets the scale relating density variation to geometric variation. In particular, since Gaussian smoothing suppresses high-frequency components of $D^2 p_\sigma$, it suppresses the fine-scale variation of the shape operator and hence removes small-scale geometric structure from the level set.

It is important to distinguish this suppression of fine-scale variation from a pointwise reduction in curvature. In general, Gaussian smoothing does not imply that the principal curvatures of a fixed level set decrease monotonically with σ. Indeed, the shape operator

$$S_\sigma(x) = -\frac{1}{\|\nabla p_\sigma(x)\|} P_T D^2 p_\sigma(x) P_T, \qquad x \in \mathcal{M}_c^\sigma, \tag{22}$$

depends on both $D^2 p_\sigma$ and ∇p_σ, and is evaluated on the moving level set $\mathcal{M}_c^\sigma$. Although Gaussian smoothing suppresses high-frequency components of $D^2 p_\sigma$, it simultaneously changes ∇p_σ and the location of the level set itself. Hence there is no general pointwise monotonicity of S_σ or of its eigenvalues.

Moreover, the topology of $\mathcal{M}_c^\sigma$ can change with σ. Such a change occurs when the level c passes through a critical point, i.e., when

$$p_\sigma(x) = c, \qquad \nabla p_\sigma(x) = 0. \tag{23}$$

For example, two components of a superlevel set may merge through a neck as σ increases. Thus, while smoothing removes fine-scale spatial variation from the density and its derivatives, it should not be interpreted as a monotone decrease of the curvature of every individual level set. The rigorous statement is instead that Gaussian smoothing preferentially suppresses spatial frequencies at scales smaller than σ, and therefore suppresses fine-scale geometric variation encoded by the derivatives of the density.

More generally, adding Gaussian noise monotonically suppresses all L^2 Sobolev energies of the smoothed density. Let

$$p_\sigma = G_\sigma * p^\star, \qquad G_\sigma(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right), \tag{24}$$

where D is the ambient dimension. Since $\partial_{\sigma^2} p_\sigma = \frac{1}{2}\Delta p_\sigma$, we have, for every integer $k \geq 0$,

$$\frac{\partial}{\partial\sigma^2} \frac{1}{2} \int_{\mathbb{R}^D} \left|D^k p_\sigma(x)\right|^2 dx = -\frac{1}{2} \int_{\mathbb{R}^D} \left|\nabla D^k p_\sigma(x)\right|^2 dx \leq 0. \tag{25}$$

Thus, increasing the noise level σ monotonically decreases the L^2 norms of all spatial derivatives of p_σ. In particular,

$$\frac{\partial}{\partial \sigma^2} \frac{1}{2} \int |\nabla p_\sigma|^2 \, dx = -\frac{1}{2} \int |\Delta p_\sigma|^2 \, dx \leq 0, \tag{26}$$

$$\frac{\partial}{\partial \sigma^2} \frac{1}{2} \int |D^2 p_\sigma|^2 \, dx = -\frac{1}{2} \int |\nabla \Delta p_\sigma|^2 \, dx \leq 0. \tag{27}$$

Equivalently, in the Fourier domain,

$$\frac{1}{2} \int_{\mathbb{R}^D} |D^k p_\sigma(x)|^2 \, dx \ \propto \ \frac{1}{2} \int_{\mathbb{R}^D} \|\xi\|^{2k} e^{-\sigma^2 \|\xi\|^2} \left| \widehat{p^\star}(\xi) \right|^2 \, d\xi. \tag{28}$$

Hence increasing σ progressively removes increasingly fine-scale structure, with spatial frequency $\|\xi\|$ suppressed according to $e^{-\sigma^2 \|\xi\|^2}$.

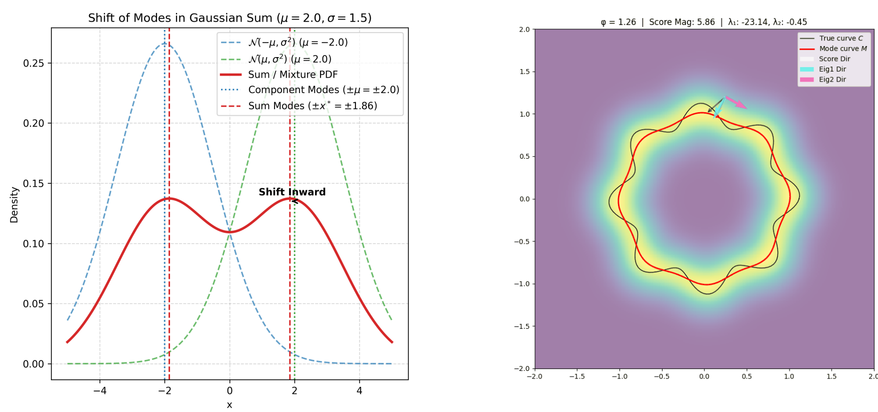
This relates to the conventional wisdom where we need more refined time steps as we approach closer to our final density to capture the higher frequency structure of density. This is particularly apparent for image and audio data. JPEG compression takes advantage of writing the image as a Fourier decomposition, and in doing so discard the high frequency terms which are not noticeable to the human eye.

5 Ridges

5.1 Motivation

An at first unintuitive aspect of the manifold hypothesis is that the lower dimensional space around which data concentrate is not necessarily, or in fact usually, the initial manifold. This is because as the amount of noise that we add increases the effective radius with which we convolve grows and includes more of our initial manifold. We see a noise density concentrating around a different space. This begs the question “what is this new space and how does it behave and evolve?”

This becomes immediately obvious when considering smoothing a pair of distinct Bernouli random variables: the modes of the sum is not the union of the modes.



(a) Gaussian smoothing moves modes in 1-D

(b) Gaussian smoothing moves the ridge in 2-D

Figure 1: Comparison of ridge structures.

With this intuition it becomes fairly easy to rediscover the definition of a ridge. It is where the normal component of our changing density is zero. But a priori we do not know which components are normal.

The resolution is to let the density define its own normal directions. At a point on a d-dimensional ridge the density behaves like a mountain ridge: it is at a local maximum in the $D - d$ directions transverse to the ridge, while along the remaining d directions it varies slowly. Transverse to the ridge the density curves sharply downward; along it, the curvature is mild. This is a statement about the Hessian. Let $g(x) = \nabla p(x)$ and $H(x) = \nabla^2 p(x)$, and order the eigenvalues of $H(x)$ as

$$\lambda_1(x) \geq \cdots \geq \lambda_d(x) \geq \lambda_{d+1}(x) \geq \cdots \geq \lambda_D(x),$$

with orthonormal eigenvectors $u_1(x), \ldots, u_D(x)$. The directions of sharpest downward curvature are those with the most negative eigenvalues, so we *define* the normal space at x to be the span of the bottom $D-d$ eigenvectors, and the tangent space to be the span of the top d. Writing $V(x) = [u_{d+1}(x) \mid \cdots \mid u_D(x)]$ for the normal frame, $L(x) = V(x)V(x)^\top$ is the projector onto the normal space. This is the point of the construction: which directions are normal is not supplied from outside, it is read off from the second-order structure of the smoothed density itself.

A ridge point is then one from which the density cannot be increased by moving normally. Equivalently, the normal component of the gradient vanishes. Define the projected gradient

$$G(x) = L(x)\, g(x),$$

and the ridge as the set where it is zero and the transverse curvature is genuinely downward:

$$R = \big\{x : \|G(x)\| = 0,\ \lambda_{d+1}(x) < 0\big\}.$$

The second condition, $\lambda_{d+1}(x) < 0$, requires the largest of the normal eigenvalues to still be negative, so that every normal direction is a direction of local descent and the point is a maximum transverse to the ridge rather than a saddle. The case $d = 0$ recovers ordinary modes: $L(x) = I$, the condition $\|G(x)\| = 0$ becomes $\nabla p(x) = 0$, and $\lambda_1(x) < 0$ asks for a local maximum. A d-ridge relaxes this by demanding a critical point only in the $D-d$ normal directions, and negative curvature only in the corresponding block.

5.2 The score concentrates on the ridge

We first record a convenient fact: the ridge of $\log p_\sigma$ and the ridge of p_σ are the same set. Since log is strictly increasing, $\nabla \log p_\sigma = p_\sigma^{-1} \nabla p_\sigma$ vanishes exactly where ∇p_σ does, and the two Hessians differ by the rank-one term $p_\sigma^{-1} \nabla p_\sigma \nabla p_\sigma^\top$, which vanishes wherever the gradient does. On the ridge, then, gradient, projected gradient and normal eigenframe all agree between p_σ and $\log p_\sigma$. We work with $\log p_\sigma$ from now on, for two reasons: it is the object the generative model actually learns, its gradient being the score; and, as Genovese et al. note, the surrogate property holds in an $O(1)$ neighbourhood of M for the log-density but only in an $O(\sigma)$ neighbourhood for the density itself.

Inside the tube of radius $O(\sigma)$ around the (rescaled) manifold, the score is overwhelmingly a normal restoring field. Genovese et al. show that $\nabla \log p_\sigma(x) = -\frac{1}{\sigma^2}\big((x - \hat{x}) + O(K_\sigma^2)\big)$, where $\hat{x} = \pi_M(x)$ is the projection onto M and

$$K_\sigma = \sqrt{2\sigma^2 \log\!\big(\sigma^{-(A+D)}\big)} = O\!\left(\sigma \sqrt{\log \tfrac{1}{\sigma}}\right).$$

The normal component has size $O(\sigma^{-1})$ while the tangential component is $O(1)$: as $\sigma \to 0$ the score is, to leading order, a stiff spring pulling transversally inward. The only question is where the spring is anchored.

The answer is not M. Write the score as a first-order expansion about the manifold projection $m = \pi_M(x)$ and, alternatively, about the ridge projection $r = \pi_{R_\sigma}(x)$:

$$\nabla \log p_\sigma(x) = \nabla_M \log p_\sigma(m) - \frac{1}{\sigma^2}(x-m) + O(K_\sigma^2) + o\left(\frac{\|x-m\|}{\sigma^2}\right), \tag{29}$$

$$\nabla \log p_\sigma(x) = \nabla_{R_\sigma} \log p_\sigma(r) - \frac{1}{\sigma^2}(x-r) + o\left(\frac{\|x-r\|}{\sigma^2}\right). \tag{30}$$

The two expansions are identical in form but for one term. Expanding about the manifold leaves a residual $O(K_\sigma^2)$; expanding about the ridge does not. This residual is not a linear-in-displacement error that dies as $x \to m$ – those are collected in the $o(\|x-m\|/\sigma^2)$ term. It is a *constant* normal offset of magnitude $K_\sigma^2/\sigma^2 = O(\log \frac{1}{\sigma})$, present even on M itself. In other words, the normal restoring field does not vanish on the manifold; its zero has been pushed off M by exactly $O(K_\sigma^2)$ in the normal direction. That displaced zero is the ridge, and $O(K_\sigma^2)$ is precisely the Hausdorff bias $\mathrm{Haus}(M, R_\sigma) = O(\sigma^2 \log \frac{1}{\sigma})$ of Theorem 7. The manifold is where the data lived; the ridge is where the smoothed score believes it lives.

Projecting (30) onto the normal space N_r at the ridge point makes this exact. The intrinsic gradient $\nabla_{R_\sigma} \log p_\sigma(r)$ lies in $T_r R_\sigma$ and is annihilated by $P_N(r)$, leaving

$$P_N(r)\, \nabla \log p_\sigma(x) = -\frac{1}{\sigma^2}(x-r) + o\left(\frac{\|x-r\|}{\sigma^2}\right).$$

The normal score is a pure linear restoring field toward the ridge, with slope $-\sigma^{-2}$ and *no* curvature remainder. This is the precise content of the slogan "the score points at the ridge": the ridge is the exact zero set of the leading normal score, and every nearby point is pulled back toward it at rate σ^{-2}. It also closes the loop with the definition of the previous subsection. There the ridge was declared to be the locus where the normal component of the smoothed gradient vanishes; here we see that this locus is not an artefact of the definition but the genuine attractor of the score's dominant term, displaced from the true manifold by a curvature bias that the model has no way to remove.

5.3 Failings of the Ridge

The objects the ridge is built from are as smooth as anything in this theory. For $t > 0$ the density p_t is a Gaussian convolution, hence real-analytic; the posterior mean m_t, the covariance field A_t, and the Hessian $\nabla^2 \log p_t = (A_t - I)/t$ are all analytic in y, and their eigenvalues $\lambda_1(y) \geq \cdots \geq \lambda_D(y)$ vary continuously everywhere. Nothing about the smoothed data is fragile. Yet the ridge set R_t is not a continuous functional of the data, and it is worth being exact about where the discontinuity enters.

It enters through the one operation the definition adds on top of the smooth field: the selection of a $d/(D-d)$ eigenspace split. Eigenvalues of a symmetric matrix are continuous, but the assignment of a matrix to the span of its bottom

$D-d$ eigenvectors is continuous only where those eigenvectors are separated from the rest of the spectrum by a gap. The controlling quantity is $\lambda_d(y) - \lambda_{d+1}(y)$, equivalently $a_d(y) - a_{d+1}(y)$. By the Davis–Kahan theorem the normal frame $P_N(y)$ is Lipschitz in $H(y)$ with constant proportional to the inverse gap, and this bound is vacuous where the gap closes. There the normal space $N(y)$ rotates discontinuously even though $H(y)$ moves analytically through the crossing.

There are in fact two independent failure modes, corresponding to the two clauses of the definition. Where the gap $a_d - a_{d+1} \to 0$, the projector P_N is ill-defined and the first ridge condition $\|P_N \nabla \log p_t\| = 0$ loses meaning: the frame that decides which directions are "normal" is spinning. Separately, where the threshold $a_{d+1}(y) = 1$ is crossed, the second condition flips on or off, and R_t gains or loses an entire component; its Hausdorff distance to its neighbours jumps and its homotopy type can change. This is not a defect that Genovese et al. overlooked. Their stability theorem bounds $\mathrm{Haus}(R, \hat{R})$ by a quantity of order ψ/β, where β is an assumed lower bound on precisely this eigengap; the hypothesis that the failure does not occur is built into the result. The instability is therefore a property of the map $(m_t, A_t) \mapsto R_t$, not of the distribution it is applied to. One takes a smooth field and passes it through a hard threshold, and thresholds are discontinuous.

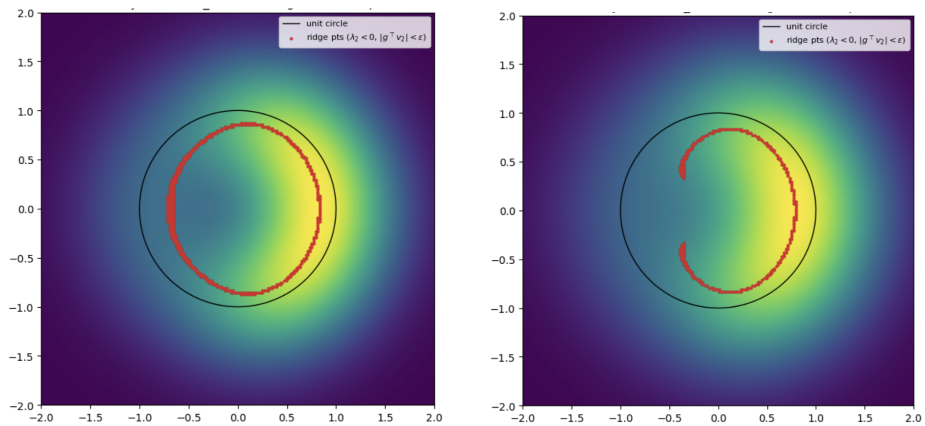
A second, more basic problem is that the split requires the integer d to be fixed in advance, and there is no consistent choice. The dimension is an input to the eigenspace selection, yet the natural scale-dependent notion of dimension, $d_{\text{eff}}(t) = \mathbb{E}\,\mathrm{tr}\,A_t$, is neither an integer nor constant in t: it interpolates between the intrinsic dimension at small noise and zero at large noise, and it overshoots in between (Chapter 6). At the noise levels a sampler actually visits, the smoothing radius is comparable to the reach, the tube has swallowed the manifold, and there is no meaningful partition of directions into d tangent and $D-d$ normal ones to be found. The ridge construction demands a decision — how many normal directions? — that the geometry at that scale simply does not answer. The posterior covariance field is under no such obligation: it reports the whole spectrum $a_1, \ldots, a_D$ and lets the number of near-tangent directions be whatever it is, at whatever scale, without a threshold and without a prior commitment to d.

Consider a distribution on $\mathbb{S}^1 \subset \mathbb{R}^2$ which is a mixture of 2 uniform densities on 2 disjoint semicircles. We let the mixture be again in a 1:3 ratio.

There are two behaviours which we see in the evolution of the ridge

Impact of Density In figure 2a we see that the less dense left half of the circle is moving towards the centre at a much faster rate that the denser right half. This is expected since the higher density areas are more robust to smoothing, since there is more mass there.

Topology Change Similar to the 1-d case, in figure 2b we see that the ridge is no longer topologically a circle. So it is not just the geometry which is changing over time but also the topology. In the case where the density is uniform over



(a) Ridge of a circle with $\sigma = 0.54$

(b) Ridge of a circle with $\sigma = 0.58$

Figure 2: Comparison of circle ridges at different noise levels σ.

the entire circle this occurs when the ridge shrinks to a point, but here it splits and then changes abruptly.

6 Posterior Covariance Field

Here we introduce the ultimate object we analyse: posterior covariance fields.

Definition 6.1. Let $Y_t = X + \sqrt{t}Z$, $m_t(y) := \mathbb{E}\left[X \mid Y_t = y\right]$, and

$$A_t(y) := \frac{1}{t}\operatorname{Cov}\left(X \mid Y_t = y\right).$$

We call the weighted object $\mathfrak{F}_t := \operatorname{Law}\left(Y_t, m_t\left(Y_t\right), A_t\left(Y_t\right)\right)$ the *posterior covariance field*.

Note: while we are also including the posterior mean $m_t(y)$ in our definition, we keep the name for simplicity. The posterior mean m_t and by extension the score s_t is the object whose regularity we want to understand, so its inclusion is natural.

There are a few reasons why it makes sense to look at this object. First, it is not plagued by the issues we encountered in the analysis of ridges or level sets in section 5 and section 4. These objects suffered from being too geometric for our use case, where we care about noise scales much greater than 0. We need a more analytic quantity, and the posterior covariance field is defined for all noise scales $t > 0$. Second, the posterior covariance captures the local geometry of the score. The posterior covariance corresponds to the Jacobian of the score. This link is given in section 2. Here we normalise the posterior covariance by t to account for the scale of noise added. Third, here we deal with expectations, rather than L^∞-type quantities like the reach τ_M. If there exists a region of the density which is geometrically complex and as a consequence has poor regularity, but which carries almost no mass, we should not let it dramatically change our understanding of the behaviour of the score.

6.1 Three functionals of the spectrum

All three quantities below are moments of the same object: the spectral measure of A_t, averaged over $y \sim p_t$. Write $a_1(y), \dots, a_D(y)$ for the eigenvalues of $A_t(y)$ and let $\mu_t := \operatorname{Law}\big(a_I(Y_t)\big)$ with I uniform on $\{1, \dots, D\}$, so that $\mathbb{E}\sum_i f(a_i) = D\int f\, d\mu_t$ for any f.

6.1.1 Effective dimension

Definition 6.2.

$$d_{\text{eff}}(t) \;:=\; \mathbb{E}\big[\operatorname{tr} A_t(Y_t)\big] \;=\; \mathbb{E}\sum_i a_i. \tag{31}$$

Proposition 6.3 (Exact MMSE identity)**.** *Let* $\operatorname{mmse}(t) := \mathbb{E}\big|X - m_t(Y_t)\big|^2$. *Then*

$$d_{\text{eff}}(t) \;=\; \frac{\operatorname{mmse}(t)}{t}. \tag{32}$$

Proof. $\operatorname{tr}\operatorname{Cov}(X \mid Y_t) = \mathbb{E}\big[|X - m_t(Y_t)|^2 \mid Y_t\big]$; take expectations and divide by t. $\square$

Identity (32) is exact, not asymptotic, and has two consequences of practical importance. First, d_{eff} is dimensionless: it is residual posterior uncertainty measured in units of injected noise variance. Second, $\text{mmse}(t)$ is precisely the population value of the ε-prediction (or x_0-prediction) denoising loss at noise level t, so d_{eff} requires no computation beyond the training objective already being evaluated.

Link to dimension. An orthogonal projector of rank k has spectrum $\{1^{(k)}, 0^{(D-k)}\}$ and trace k. In the ideal manifold limit $A_t \approx P_{T_x\mathcal{M}}$ and hence $d_{\text{eff}}(t) \approx d$. Equivalently, via (32): a d-dimensional manifold leaves $O(t)$ posterior uncertainty in each of d tangent directions and $o(t)$ in the normal directions, so $\text{mmse} \approx d \times t$. Each eigenvalue may be read as a fractional dimension: $a_i = 0.4$ contributes 0.4 effective directions.

Failure mode. d_{eff} is a first moment and therefore cannot distinguish $\text{diag}(1,1,1,0,0,0)$ from $\text{diag}(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2})$: both have trace 3. It also over-counts when $a_i > 1$, in which case the "fractional dimension" reading breaks down entirely. The next two functionals repair exactly these two deficiencies.

6.1.2 Distance to projector structure

Definition 6.4.

$$\Delta_{\text{proj}}(t) \;:=\; \mathbb{E}\sum_i \min\{a_i^2,\ (a_i-1)^2\}. \tag{33}$$

The summand is the squared distance from a_i to the set $\{0,1\}$, and the two branches exchange at $a_i = \frac{1}{2}$:

$$\min\{a^2,(a-1)^2\} = \begin{cases} a^2, & a \le \frac{1}{2} \quad \text{(read as normal-like)}, \\ (a-1)^2, & a \ge \frac{1}{2} \quad \text{(read as tangent-like)}. \end{cases} \tag{34}$$

Proposition 6.5. *Let $\mathcal{P} = \{P \in \mathbb{R}^{D\times D} : P = P^\top = P^2\}$ be the set of orthogonal projectors and let A be symmetric with eigenvalues a_i. Then*

$$\min_{P\in\mathcal{P}} \|A-P\|_F^2 \;=\; \sum_i \min\{a_i^2,(a_i-1)^2\}, \qquad \textit{hence} \qquad \Delta_{\text{proj}}(t) = \mathbb{E}\big[\text{dist}_F^2\big(A_t(Y_t),\mathcal{P}\big)\big]. \tag{35}$$

Proof. $\mathcal{P}$ is invariant under conjugation by orthogonal matrices, and for symmetric A the Frobenius distance to a conjugation-invariant set is minimized by a P sharing the eigenvectors of A (a consequence of the Hoffman–Wielandt inequality). It then remains to round each a_i to the nearer of 0 or 1. □

Thus Δ_{proj} answers: *how far, on average, is the soft posterior geometry from being a genuine tangent-space projector?* It vanishes exactly when $A_t(Y_t)$ is a projector almost surely.

Remark 6.6 (Interpretation as a resolution probe). For data concentrated near a manifold but with normal thickness δ, the normal eigenvalues behave as $a_n \approx \delta^2/(\delta^2 + t)$, which passes through $\frac{1}{2}$ at $t \approx \delta^2$ and contributes $\frac{1}{4}$ to Δ_{proj} there. Consequently $\Delta_{\text{proj}}(t)$ peaks at the scale at which the manifold hypothesis stops being exactly true, and its peak location is a readout of that scale. Δ_{proj} is best regarded as a scale-resolved test of manifold-likeness rather than a single verdict.

6.1.3 Ambiguity

Definition 6.7. With $(x)_+ := \max\{x, 0\}$,

$$\mathcal{A}(t) \; := \; \mathbb{E} \sum_i \big(a_i - 1\big)_+^2. \tag{36}$$

The motivation is dynamical rather than geometric. $a_i > 1$ occurs if and only if the reverse probability-flow ODE is locally expansive along v_i. The threshold is the separatrix at which posterior variance exceeds the injected noise variance:

$$a_i = \frac{\text{posterior variance along } v_i}{\text{noise variance } t} \qquad \begin{cases} < 1: & \text{denoising, contraction,} \\ = 1: & \text{neutral,} \\ > 1: & \text{amplification, expansion.} \end{cases} \tag{37}$$

The three design choices are then: threshold at 1 (only expansion is penalized), measure the *excess* $a_i - 1$ (the neutral point is 1, not 0), and square it (making the functional comparable to a squared Frobenius norm and penalizing strong expansion superlinearly).

Link to ambiguity. Suppose the posterior at y is approximately $\frac{1}{2}\mathcal{N}(\mu_1, \Sigma) + \frac{1}{2}\mathcal{N}(\mu_2, \Sigma)$, corresponding to two competing clean configurations. Then

$$\operatorname{Cov}(X \mid Y_t = y) \; = \; \Sigma + \tfrac{1}{4}(\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top, \tag{38}$$

and the between-mode term can exceed tI along $\mu_1 - \mu_2$, giving $a_i > 1$. So $a_i > 1$ signals that the noisy observation fails to identify a single locally stable latent configuration.

Remark 6.8 (Scope of the interpretation). Strictly, $a_i > 1$ is a diagnostic that posterior variance exceeds the noise scale. Multimodality is the typical, but not the only, mechanism: heavy-tailed unimodal p_0 (e.g. Laplace, Student-t) also produces $a_i > 1$. "Ambiguity" is the geometric reading of the phenomenon, not a theorem.

Remark 6.9 (Mean versus tail). $\mathcal{A}$ is an average over $y \sim p_t$ and can be small while $\sup_y a_{\max}(y)$ is large, if the expansive region carries little mass. In practice one should report both $\mathcal{A}(t)$ and a high quantile of $(a_{\max} - 1)_+$, since sampler failure is driven by trajectories that enter the expansive region, not by its measure.

6.2 MMSE-driven discretization

We started by looking at (Potaptchik et al., 2024).

We consider the additive Gaussian perturbation

$$X_t = X_0 + \sqrt{t}\,Z, \qquad Z \sim \mathcal{N}(0, I_D), \tag{39}$$

so that, in the notation of the diffusion model,

$$c_t = 1, \qquad \sigma_t^2 = t. \tag{40}$$

Let

$$m_t(x) := \mathbb{E}[X_0 \mid X_t = x] \tag{41}$$

denote the posterior mean and define the minimum mean-squared error

$$\mathrm{MMSE}(t) := \mathbb{E}\left[\|X_0 - m_t(X_t)\|^2\right]. \tag{42}$$

Following the effective-dimension viewpoint, we write

$$d_{\mathrm{eff}}(t) := \frac{\mathrm{MMSE}(t)}{t}, \tag{43}$$

and hence

$$\mathrm{MMSE}(t) = t\, d_{\mathrm{eff}}(t). \tag{44}$$

Discretization error. For additive Gaussian noise, Tweedie's formula gives

$$s_t(x) = \frac{m_t(x) - x}{t}. \tag{45}$$

The same martingale argument used to control the score variation in the discretized reverse process then yields, for two noise levels $t < t'$,

$$\begin{aligned} &\mathbb{E}\left\|s_t(X_t \mid t', X_{t'}) - s_t(X_t)\right\|^2 \\ &\quad = \frac{1}{t^2}\left(\mathrm{MMSE}(t') - \mathrm{MMSE}(t)\right), \end{aligned} \tag{46}$$

with the sign understood according to the direction in which the diffusion trajectory is traversed. Thus, for a sufficiently fine discretization with step size

$$\Delta t_k := t_{k+1} - t_k, \tag{47}$$

the contribution of the k-th interval is, up to higher-order terms,

$$\mathcal{E}_k \;\asymp\; \frac{\left|\mathrm{MMSE}'(t_k)\right|}{t_k^2}(\Delta t_k)^2. \tag{48}$$

Consequently, the total discretization error is approximated by

$$\mathcal{E}_{\mathrm{disc}} \;\asymp\; \sum_{k=0}^{K-1} a(t_k)(\Delta t_k)^2, \qquad a(t) := \frac{|\,\mathrm{MMSE}'(t)|}{t^2}. \tag{49}$$

Optimal allocation of time steps. We now choose the discretization points to minimize (49) subject to a fixed number K of steps. Introduce the local step density

$$n(t) := \frac{1}{\Delta t(t)}. \tag{50}$$

In the continuum limit,

$$K \approx \int n(t)\, dt, \tag{51}$$

while

$$\begin{aligned} \mathcal{E}_{\text{disc}} &\approx \int a(t)\Delta t(t)\, dt \\ &= \int \frac{a(t)}{n(t)}\, dt. \end{aligned} \tag{52}$$

Minimizing this functional subject to $\int n(t)\, dt = K$ gives

$$n(t) \propto \sqrt{a(t)}, \tag{53}$$

and therefore

$$\Delta t(t) \propto \frac{1}{\sqrt{a(t)}} = \frac{t}{\sqrt{|\,\text{MMSE}'(t)|}}. \tag{54}$$

Finally, using (44),

$$\text{MMSE}'(t) = \frac{d}{dt}\left[t\, d_{\text{eff}}(t)\right] = d_{\text{eff}}(t) + t\, d'_{\text{eff}}(t), \tag{55}$$

so the resulting MMSE-adaptive discretization is

$$\boxed{\Delta t(t) \propto \frac{t}{\sqrt{|d_{\text{eff}}(t) + t\, d'_{\text{eff}}(t)|}}}. \tag{56}$$

This prescription allocates more discretization points to regions in which the posterior uncertainty, measured by the MMSE, changes rapidly, while accounting for the t^{-2} amplification of score errors at small noise scales.

6.3 Examples

6.4 Summary

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